RIGHT ADJOINT F^! TO LOWER SHRIEK FOR LOCALLY QUASI-FINITE MORPHISMS AND ITS VALUE ON SKYSCRAPER SHEAVES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Disclosure: two soft spots, both mechanical identifications over displayed formulas rather than omitted verifications. (1) After the two displayed Hom computations (proof lines 2429-2443) the author writes 'We leave it to the reader to identify these as functors of F' (proof line 2444): the identification is the standard adjunction between direct sums and products of abelian groups, with both sides visibly functorial in F through the displayed expressions. (2) The module-level adjunction in part (3) is concluded by 'A straightforward computation shows that they are adjoints' (proof line 2451): module objects are abelian sheaves with a map Lambda -> End, and the adjunction at the abelian level from part (1) transports along this description. No verbatim 'we omit' occurs in the proof.

Allowed standard prerequisites (editorial).

- Partially defined adjoints, dual form (Homology, Lemma homology-lemma-partially-defined-adjoint, canonical tag 0793). cited at proof lines 2398-2400. Produces the right adjoint f^! once representability on the class E is established.
- Exactness of stalks (Etale Cohomology, Theorem etale-cohomology-theorem-exactness-stalks, canonical tag 03PU). cited at proof lines 2403-2406. Gives the injection of a sheaf into the product of its skyscraper sheaves (part (a)).
- Stalk description of locally quasi-finite lower shriek (Lemma lemma-lqf-f-shriek-stalk, canonical tag 0F5F). cited at proof lines 2434-2435. Identifies the stalk of f_!F with the direct sum over the fibre, from which the representing object is computed.

Source reference. Original theorem 03PU: sources/stacks-chow/tags.txt, source lines 4871–4871. Cited at proof lines 2403-2406: the injection of a sheaf into the product of its skyscraper sheaves in part (a).

Source reference. Original lemma 0793: sources/stacks-chow/tags.txt, source lines 9258–9258. Cited at proof lines 2398-2400: the dual form produces the right adjoint f^! once representability on E is shown.

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Source reference. Original lemma 0F5F: sources/stacks-chow/tags.txt, source lines 18935–18935. Cited at proof lines 2434-2435: the stalk of f_!F is the direct sum over the fibre, the input to the representability computation.

Lemma 0.2. *Let $f : X \to Y$ be a locally quasi-finite morphism of schemes.*

1. *The functor $f_! : Ab(X_{étale}) \to Ab(Y_{étale})$ has a right adjoint $f^! : Ab(Y_{étale}) \to Ab(X_{étale})$.*
2. *We have $f^!(\overline{y}_* A) = \prod_{f(\overline{x})=\overline{y}} \overline{x}_* A$.*
3. *If Λ is a ring, then the functor $f_! : Mod(X_{étale}, \Lambda) \to Mod(Y_{étale}, \Lambda)$ has a right adjoint $f^! : Mod(Y_{étale}, \Lambda) \to Mod(X_{étale}, \Lambda)$ which agrees with $f^!$ on underlying abelian sheaves.*

Proof. Proof of (1). Let $E \subset \mathrm{Ob}(Ab(Y_{étale}))$ be the class consisting of products of skyscraper sheaves. We claim that

(a) every $\mathcal{G}$ in $Ab(Y_{étale})$ is a subsheaf of an element of E, and
(b) for every $\mathcal{G} \in E$ there exists an object $\mathcal{H}$ of $Ab(X_{étale})$ such that $\mathrm{Hom}(f_!\mathcal{F}, \mathcal{G}) = \mathrm{Hom}(\mathcal{F}, \mathcal{H})$ functorially in $\mathcal{F}$.

Once the claim has been verified, the dual of Homology, Lemma 0793 produces the adjoint functor $f^!$.

Part (a) is true because we can map $\mathcal{G}$ to the sheaf $\prod \overline{y}_* \mathcal{G}_{\overline{y}}$ where the product is over all geometric points of Y. This is an injection by Étale Cohomology, Theorem 03PU. (This is the first step in the Godement resolution when done in the setting of abelian sheaves on topological spaces.)

Part (b) and part (2) of the lemma can be seen as follows. Suppose that $\mathcal{G} = \prod \overline{y}_* A_{\overline{y}}$ for some abelian groups $A_{\overline{y}}$. Then

$$\mathrm{Hom}(f_!\mathcal{F}, \mathcal{G}) = \prod \mathrm{Hom}(f_!\mathcal{F}, \overline{y}_* A_{\overline{y}})$$

Thus it suffices to find abelian sheaves $\mathcal{H}_{\overline{y}}$ on $X_{étale}$ representing the functors $\mathcal{F} \mapsto \mathrm{Hom}(f_!\mathcal{F}, \overline{y}_* A_{\overline{y}})$ and to take $\mathcal{H} = \prod \mathcal{H}_{\overline{y}}$. This reduces us to the case $\mathcal{H} = \overline{y}_* A$ for some fixed geometric point $\overline{y} : \mathrm{Spec}(k) \to Y$ and some fixed abelian group A. We claim that in this case $\mathcal{H} = \prod_{f(\overline{x})=\overline{y}} \overline{x}_* A$ works. This will finish the proof of parts (1) and (2) of the lemma. Namely, we have

$$\mathrm{Hom}(f_!\mathcal{F}, \overline{y}_* A) = \mathrm{Hom}_{Ab}((f_!\mathcal{F})_{\overline{y}}, A) = \mathrm{Hom}_{Ab}(\bigoplus\nolimits_{f(\overline{x})=\overline{y}} \mathcal{F}_{\overline{x}}, A)$$

by the description of stalks in Lemma 0F5F on the one hand and on the other hand we have

$$\mathrm{Hom}(\mathcal{F}, \mathcal{H}) = \prod\nolimits_{f(\overline{x})=\overline{y}} \mathrm{Hom}(\mathcal{F}, \overline{x}_* A) = \prod\nolimits_{f(\overline{x})=\overline{y}} \mathrm{Hom}_{Ab}(\mathcal{F}_{\overline{x}}, A)$$

We leave it to the reader to identify these as functors of $\mathcal{F}$.

Proof of part (3). Observe that an object $Mod(X_{étale}, \Lambda)$ is the same thing as an object $\mathcal{F}$ of $Ab(X_{étale})$ together with a map $\Lambda \to \mathrm{End}(\mathcal{F})$. Hence the functors $f_!$ and $f^!$ in (1) define functors $f_!$ and $f^!$ as in (3). A straightforward computation shows that they are adjoints. □